

CAPSTONE PROJECT · DATA ANALYTICS WITH GENAI

Black Friday Sales Analysis

Turning 550,068 transactions into a data-driven retail strategy

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BUSINESS UNDERSTANDING

BLACK FRIDAY SALES ANALYSIS



Business Context

Black Friday is a single high-stakes sales window: millions of shoppers, thousands of products, and immense pressure on pricing, promotions, and inventory — for a retailer spanning both metro and non-metro cities.



Key Challenges

Fragmented customer behavior across age, occupation, and marital status · Uneven city-level purchasing power · Unpredictable product-category demand · Unclear return on promotional spend.



Analytical Objectives

Identify high-value and underpenetrated customer segments · Quantify how demographics shape spend · Surface top-performing categories and cross-sell opportunities · Recommend a data-backed retention strategy.



DATASET OVERVIEW

BLACK FRIDAY SALES ANALYSIS

550,068

Transactions

12

Columns

5,891

Unique Customers

3,631

Unique Products

What the data captures

- Customer demographics — gender, age band, occupation code, marital status
- City context — city category (A/B/C) and years in current city
- Product hierarchy — up to 3 product-category levels per transaction
- Purchase amount — the transaction-level target variable



DATA QUALITY & CLEANING

BLACK FRIDAY SALES ANALYSIS



Missing Values

Product_Category_2 was missing in ~32% of rows and Product_Category_3 in ~70% — expected, since not every product carries a secondary or tertiary category.



Data Validation

Excel conditional formatting was used to check key fields (User_ID, Product_ID, Purchase), flag Purchase outliers, and confirm categorical columns had no inconsistent entries.



Treatment of Category 2 & 3

Blanks were deliberately converted to true NULLs during SQL load (via NULLIF) rather than imputed — preserving the honest signal that no category was recorded.



Other Cleaning Performed

Verified zero duplicate rows, confirmed all Age / Gender / City / Tenure values were clean, and validated the final row count (550,068) after load.



KPI OVERVIEW

BLACK FRIDAY SALES ANALYSIS

₹5.09B

Total Sales

₹9,264

Average Purchase

₹865.02K

Avg. Purchase / Customer

~100%

Repeat Purchase Rate

Cat. 1, 5, 8

Category Penetration (top)

City B

Highest City Contribution

35.7%

High-Value Segment Share

Seven KPIs defined in Step 1 were calculated end-to-end and tracked live on the Power BI Executive Dashboard.



CUSTOMER ANALYSIS

BLACK FRIDAY SALES ANALYSIS



Gender

Male customers spend more: ₹9,437.53 vs. ₹8,734.57 for female customers — a statistically significant gap (t-test, $p < 0.001$).



Age

26–35 is the leading age group by total spend. Age has a statistically significant but comparatively modest effect (ANOVA $F = 40.6$).



Marital Status

No meaningful difference: Single ₹9,265.91 vs. Married ₹9,261.17 ($p = 0.731$) — an initial hypothesis this project disproved.



Occupation

Correlation with purchase amount is negligible ($r = 0.02$) despite a tiny p-value — not a practically useful targeting variable.



Customer Segments: High-Value Loyal 35.7% · Regular 33.1% · Loyal 31.2% — no meaningful one-time-buyer or deal-seeker segment emerged.



PRODUCT ANALYSIS

BLACK FRIDAY SALES ANALYSIS



Top Categories

Categories 1, 5, and 8 show by far the highest transaction volume — the clearest “penetration” leaders across the customer base.



Underperforming Categories

Product_Category_1 correlates negatively with purchase value ($r = -0.34$): higher-numbered categories skew toward lower-value items.



Product / Category Affinity

Customers routinely purchase across multiple category levels per visit, reflecting broad cross-category browsing rather than single-item trips.



Cross-Selling Opportunities

A single hero product (P00025442) generated ₹27.99M alone — bundling adjacent SKUs within categories 1, 5, and 8 is the clearest cross-sell lever.



GEOGRAPHICAL ANALYSIS

BLACK FRIDAY SALES ANALYSIS

Sum of Purchase by City Category



City B

Volume Leader

Highest total revenue — driven by the largest active customer base rather than the biggest average ticket.

City C

Value Leader

Highest average purchase per transaction — a smaller but higher-spending customer base (ANOVA-confirmed).



Statistical takeaway: City Category is the strongest driver of purchase behavior of every variable tested (ANOVA $F = 1130.7$) — over 27× the effect size of Age.



STATISTICAL ANALYSIS

BLACK FRIDAY SALES ANALYSIS

Hypothesis / Test	Method	Result
Metro (City A) customers spend more	ANOVA	Not supported — City C leads
Married customers show greater category breadth	T-test	Not supported ($p = 0.731$)
Purchase amount varies by gender	T-test	Supported ($p < 0.001$)
Purchase varies significantly across city categories	ANOVA	Strongly supported ($F = 1130.7$)
Occupation / city tenure predict spend	Pearson r	Negligible effect ($r < 0.03$)



Key Conclusion: City Category and Gender are the two statistically and practically significant drivers of purchase behavior. Marital Status, Occupation, and City Tenure remain statistically “significant” only because of the huge sample size — their real-world effect is negligible.



POWER BI DASHBOARDS

BLACK FRIDAY SALES ANALYSIS

Executive Dashboard



Total Sales, Avg. Purchase/Customer, Top Customers & Products, City-wise revenue — with live Age / Occupation / Gender / Category filters.

Marketing Dashboard



Age/Gender segmentation, Repeat Purchase Rate, Category Penetration, and the Customer Segment breakdown (High-Value Loyal / Loyal / Regular).



TOP 5 BUSINESS INSIGHTS

BLACK FRIDAY SALES ANALYSIS

1

Gender significantly drives purchase value.

Male customers average ₹9,437.53/txn vs. ₹8,734.57 for female customers ($p < 0.001$).

2

City Category is the strongest — and most counterintuitive — driver of spend.

City C leads on average purchase; City B leads on total revenue via volume, not “metro” City A.

3

This customer base is exceptionally loyal.

~93 transactions per customer on average; effectively no one-time-buyer segment exists.

4

A small set of products and categories drives disproportionate revenue.

One product alone (P00025442) generated ₹27.99M; Categories 1, 5, 8 dominate volume.

5

Marital status does not meaningfully affect spending.

An initial hypothesis, disproved: Single vs. Married spend is statistically identical ($p = 0.731$).



STRATEGIC RECOMMENDATIONS

BLACK FRIDAY SALES ANALYSIS

1

City-specific strategy: Volume vs. Value

Run acquisition-focused campaigns in City B and premium/personalized offers in City C rather than one national campaign.

Expected impact: Higher ROI per marketing rupee spent.

2

Tiered loyalty program for High-Value Loyal customers

Build exclusive perks (early access, bundles) around the 35.7% segment that already drives the most repeat, high-value engagement.

Expected impact: Stronger retention & lifetime value.

3

Concentrate inventory on hero categories & products

Prioritize stock and promotional spend on Categories 1, 5, 8 and top SKUs like P00025442 to avoid stockouts during peak demand.

Expected impact: Fewer stockouts, larger basket size.

4

Reallocate budget away from marital-status targeting

Shift spend toward gender-based and city-based targeting, the two variables with proven statistical and practical significance.

Expected impact: More efficient campaign spend.



CONCLUSION

This project carried 550,068 raw transactions through a complete analytics pipeline — Excel validation, a normalized MySQL schema, Python-based statistical testing, and two interactive Power BI dashboards — to produce insights a retailer can act on immediately.

- City Category and Gender are the confirmed, statistically validated drivers of spend
- An exceptionally loyal customer base with an untapped high-value segment (35.7%)
- A small set of hero categories and products carrying disproportionate revenue
- Four concrete recommendations translating analysis directly into retail strategy

Thank you.